Alemu Sisay Nigru

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Nationality: Ethiopian

Professional Summary

AI Research Scientist and PhD candidate in Artificial Intelligence in Medicine with 3+ years of experience developing machine learning and deep learning solutions for medical imaging and clinical decision support. Specialized in spine MRI analysis, multimodal data integration, and risk prediction using imaging and patient-reported outcomes. Strong background in Python-based AI development, transformer architectures, and deployment of AI models in real-world healthcare settings. Proven ability to collaborate across clinical, academic, and international environments (Italy; USA; Ethiopia).

Technical Skills

Programming Languages: Python, C++, C#, Matlab, Java
Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn
Medical Imaging: DICOM, NIfTI, PyDICOM, Radiology AI, MRI segmentation & classification
Computer Vision: CNNs, Transformers, Image Segmentation, Feature Extraction
Data & Tools: NumPy, OpenCV, Git
Domains: Medical Imaging AI, Healthcare AI, Computer Vision, Risk Prediction
Soft Skills: Interdisciplinary collaboration, scientific communication, cross-cultural teamwork

Education

Ph.D. in Artificial Intelligence in Medicine
University of Brescia, Italy | 2022 – 2025 (Expected)
Dissertation: AI-driven clinical decision support for personalized management of low back pain.

M.Sc. in Communication Technologies and Multimedia
University of Brescia, Italy | 2019 – 2022
Final Grade: 110cum laude/110
Thesis: Deep learning–based neuroimage retrieval

B.Sc. in Electrical and Computer Engineering
University of Gondar, Ethiopia | 2012 – 2016
Final Grade: 3.89/4.00 — **Gold Medalist**
Thesis: Fingerprint-based Attendance System (FAS) using ASP.NET and C#

RESEARCH & PROFESSIONAL EXPERIENCE

Visiting PhD Researcher – Video Lab
New York University (NYU), USA | Apr 2025 – Oct 2025

- Developed deep learning models for lumbar spine pathology detection from MRI.
- Conducted training, evaluation, and benchmarking of computer vision models on clinical datasets.
- Collaborated with multidisciplinary teams on imaging pipelines and model interpretability.

Visiting PhD Researcher – Center for Digital Health and Implementation Science (CDHI)
Ethiopia | Sep 2024 – Dec 2024

- Deployed AI model APIs and contributed to frontend integration for clinical applications.
- Adapted AI tools for resource-constrained healthcare environments.
- Supported real-world validation of AI-based clinical decision support tools.

Assistant Lecturer – Department of Electrical & Computer Engineering
University of Gondar, Ethiopia | Aug 2016 – Aug 2019

- Taught undergraduate courses in C++ programming, digital logic design, circuit theory, and communication systems
- Led community-based technology transfer and capacity-building initiatives

KEY PROJECTS

- **Spine MRI Analysis with Deep Learning**
  Developed CNN and transformer-based models for segmentation and grading of lumbar spine pathologies.
- **Multimodal Risk Stratification in Spine Health**
  Integrated radiological severity with psychosocial PROMs to identify patient risk profiles and support personalized treatment strategies.
- **Brain MRI Tissue Segmentation**
  Built automated pipelines for segmentation of CSF, GM, WM, basal ganglia, and cerebrum from T1/T2 MRI scans.

PUBLICATIONS:

Peer-Reviewed

Nigru, A. S., et al. *External validation of SpineNetV2 on a comprehensive set of radiological features for grading lumbosacral disc pathologies.* **North American Spine Society Journal (NASSJ)**, 2024.

Under Review / Preprints

- **Nigru, A. S., et al.** *Toward a Clinically Integrated AI Framework for Personalized Spine Care.* SSRN preprint.
- **Nigru, A. S**, Svanera, M.., et al. *Rewiring Development: Leveraging Adult Brain Priors for Enhancing Infant Brain MRI Segmentation.* arXiv preprint.
- **Nigru, A. S**., et al. *LumbarAI: A Swin3D Transformer-Based Framework for Computer-Aided Diagnosis of Lumbar Spine Disorders from MRI (under review).*
- **Nigru, A. S**., et al. *Integrating Radiological Severity and Patient-Reported Outcomes for Risk Factor Identification in Spine Health.*

CERTIFICATIONS & TRAINING:

- Oxford Machine Learning Summer School — AI for Global Goals, University of Oxford
- Applied Data Science Lab — WorldQuant University
- Higher Diploma Program Certificate in Teaching Methods — University of Gondar
- Additional short courses in Machine Learning, Deep Learning, and AI

REFERENCES (Available upon request):

- Prof. Riccardo Leonardi, Professor @University of Brescia (riccardo.leonardi@unibs.it)
- Prof. Sergio Benini, Professor @University of Brescia (sergio.benini@unibs.it)
- Prof. Yao Wang, Professor @NYU (yw523@nyu.edu)